LAB 6 – BRANCH COVERAGE ANALYSIS AND REPORT

Experiment: Branch Coverage Analysis of Bus Transit Fare Application
Application: bus_transit_fare.py
Test File: BranchCoverageTest_PES2UG24CS570_application.py

1. Objective

To analyse the branch coverage of the Bus Transit Fare application using Python's Coverage.py tool and pytest. Additional test cases were developed to execute the different decision outcomes and control-flow branches in the application.

2. Introduction

Branch coverage measures whether the possible outcomes of decision points in a program have been executed during testing.

Unlike code coverage, which checks whether statements have been executed, branch coverage checks whether the different outcomes of conditional statements have been tested.

For this experiment, branch coverage was measured using Coverage.py with the --branch option. The aim was to achieve at least 90% branch coverage.

3. Test Case Development

Additional test cases were written to exercise the different decision branches in the Bus Transit Fare application.

The test cases covered:

- Valid and invalid manifest inputs.
- Valid and invalid booking fee conditions.
- Active and inactive routes.
- Supported and unsupported route types.
- Different passenger types and fare calculations.
- Valid and invalid commuter zones.
- Blacklisted and non-blacklisted commuters.
- Priority boarding enabled and disabled.
- Express-route blacklisting behaviour.
- Audit threshold conditions for rejected rides.
- Zero-ridership and completed manifest outcomes.

- Empty and non-empty turnstile readings.
- Consecutive turnstile failures and operational conditions.
- Turnstile jam detection and failure streak resets.

A total of 26 test cases were executed successfully.

4. Execution

The following commands were used to execute the test cases with branch measurement and generate the branch coverage report:

python -m coverage erase

python -m coverage run --branch --include="bus_transit_fare.py" -m pytest BranchCoverageTest_PES2UG24CS570_application.py

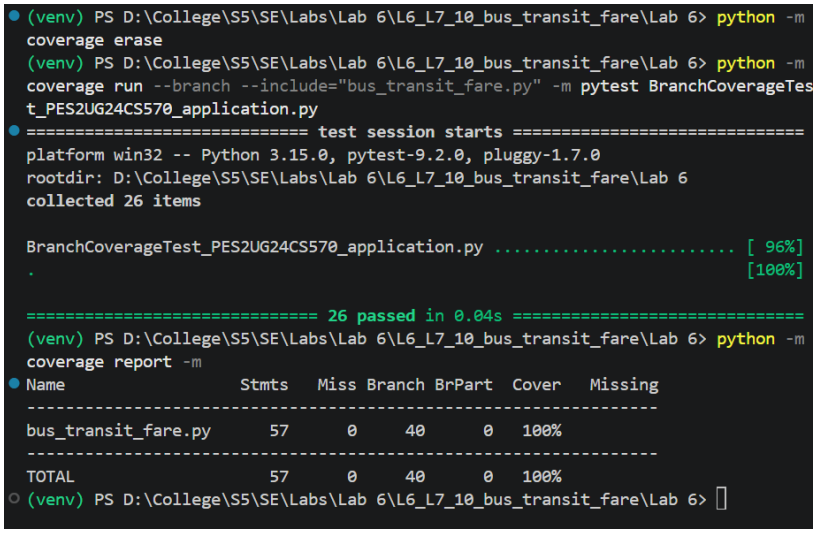
python -m coverage report -m

5. Results and Analysis

The test execution completed successfully with all 26 test cases passing.

Metric	**Result**
Total statements	57
Missing statements	0
Total branches	40
Partial branches	0
Branch coverage	100%

The coverage report showed that all 57 executable statements and all 40 branches were covered. No partial branches remained.



6. Conclusion

Branch coverage analysis was successfully performed on the Bus Transit Fare application using pytest and Coverage.py. The additional test cases exercised all identified decision branches and achieved 100% branch coverage, exceeding the required minimum of 90%. This demonstrates that both outcomes of the application's decision points were exercised during testing.